SYSTEM DESIGN DOCUMENT

1. System Overview

The Mantis Swarm System is a distributed, network-enabled cruise missile architecture designed to conduct coordinated, low-altitude strike operations at ranges up to 500 kilometers. The system is engineered for resilience, adaptability, and mission continuity in contested environments. It achieves these characteristics through a decentralized control model composed of interacting node types rather than reliance on a singular command authority.

The system is fully compatible with the Chimaera guided missile cruiser launch system, enabling rapid deployment from maritime platforms without modification to existing launch infrastructure. Once deployed, the swarm operates autonomously as a cooperative network, dynamically assigning roles and adapting to battlefield conditions in real time.

2. System Architecture

The Mantis Swarm is composed of three functional subsystems: Control Nodes, Sensor Nodes, and Strike Assets. These elements form an interconnected mesh network that enables distributed decision-making and continuous target engagement.

2.1 Control Node Subsystem

The Control Node subsystem is responsible for coordination, decision-making, and task allocation across the swarm. Unlike traditional centralized command systems, the Mantis architecture employs the **RAFT consensus protocol** to dynamically elect a **Swarm Leader** from among available control nodes.

RAFT is specifically implemented to ensure that the elimination or disruption of any single command node cannot compromise mission execution. Leadership is determined through majority voting, and authority is continuously validated through heartbeat signaling and quorum maintenance. If the current leader becomes unavailable due to failure or external interference, the remaining nodes automatically initiate a new election cycle and reconstitute leadership within operational time constraints.

Once elected, the leader node aggregates incoming sensor data, evaluates target validity, prioritizes engagements, and assigns strike assets accordingly. Importantly, leadership is not static. It is fluid and resilient, ensuring continuity even under degraded network conditions.

2.2 Sensor Node Subsystem

The Sensor Node subsystem is responsible for target detection, tracking, and continuous telemetry generation. Sensor nodes operate as distributed observers within the swarm, each capable of identifying and maintaining lock on potential targets within the operational environment.

Upon detecting a target, a sensor node begins persistent tracking and transmits telemetry to the control node network. Once a strike asset is assigned, the originating sensor node establishes a direct communication link with that strike asset. This direct linkage reduces latency and allows for precise, real-time guidance updates throughout the engagement.

Sensor nodes are designed to operate redundantly, allowing multiple nodes to track the same target. This redundancy ensures that the loss of any individual sensor does not result in target loss or mission degradation.

2.3 Strike Asset Subsystem

Strike Assets are the kinetic execution elements of the Mantis Swarm. These assets are responsible for navigating to and neutralizing designated targets. Each strike asset operates through a layered guidance model consisting of midcourse navigation, sensor-linked guidance, and terminal guidance.

After assignment by the control node leader, the strike asset receives initial routing instructions and establishes a direct connection with the sensor node responsible for target acquisition. During the midcourse phase, the strike asset follows sensor-provided updates to maintain an accurate trajectory.

At approximately 1000 meters from the target, the strike asset transitions to its terminal guidance system, which attempts to independently acquire and finalize the strike. If the terminal system successfully acquires the target, it executes the final engagement autonomously. If acquisition fails, the strike asset continues to rely on sensor-provided guidance, ensuring continuity of the attack rather than mission abort.

3. Operational Range and Flight Profile

The Mantis Swarm system is designed for low-altitude cruise operations out to 500 kilometers at a maximum speed of 486 knots. The low-altitude flight profile enhances survivability by reducing detection probability and exploiting terrain masking. The system's guidance architecture supports sustained low-level navigation while maintaining continuous communication between swarm elements.

4. Communication and Network Topology

The swarm operates as a distributed mesh network with multiple communication pathways:

- Control nodes maintain consensus synchronization using RAFT messaging.
- Sensor nodes continuously uplink target data to the control network.
- Control nodes issue tasking directives to strike assets.
- Sensor nodes maintain direct, persistent links with assigned strike assets for guidance.

This layered communication structure ensures that even if portions of the network are degraded, sufficient connectivity remains to complete the mission.

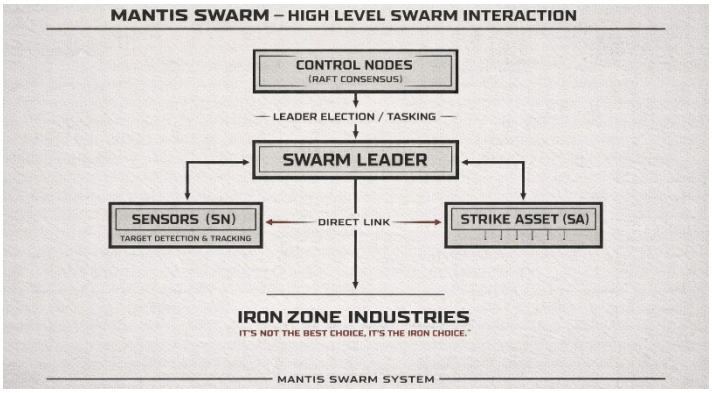
5. Fault Tolerance and Resilience

Resilience is a foundational design principle of the Mantis Swarm System. The use of the RAFT protocol eliminates single points of failure in command and control. Sensor redundancy ensures continuous target tracking, while strike assets are capable of operating under multiple guidance modes.

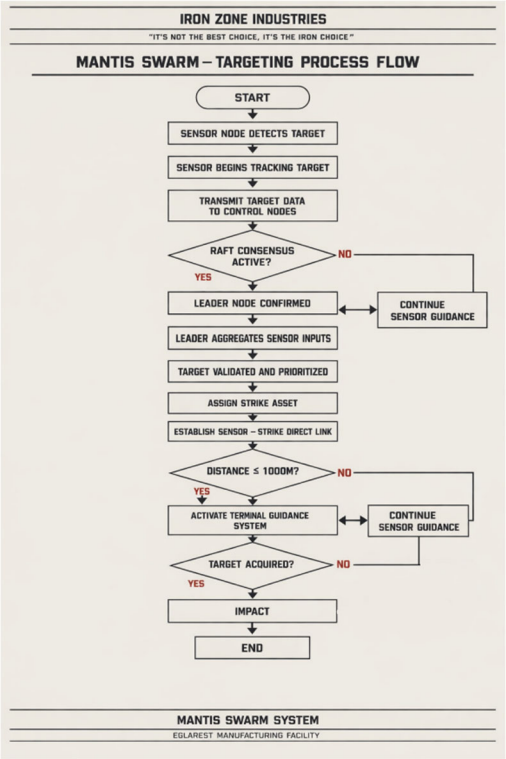
If a leader node is lost, a new leader is elected. If a sensor node is lost, another sensor can assume tracking responsibilities. If terminal guidance fails, the system reverts to sensor-based guidance. This layered redundancy ensures that mission success does not depend on any single component.

6. Targeting Process Flow

6.1 High-Level Swarm Interaction Diagram



6.2 Detailed Targeting Flow Chart



7. Design Summary

The Mantis Swarm System represents a fully distributed strike architecture designed for contested environments where traditional centralized control systems are vulnerable. By leveraging the RAFT consensus protocol, the system ensures that command authority is continuously maintained without reliance on any single node. The integration of sensor-driven targeting, direct guidance links, and layered strike logic enables high reliability and adaptability.